

Shubham Khatri

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B.Tech student in Instrumentation and Control Engineering at NSUT with a strong focus on AI/ML and full-stack development.
Built RAG systems, deep learning models, and scalable applications with measurable performance improvements. OpenCV contributor and IEEE leader with hands-on research and engineering experience.

Education

Netaji Subhas University of Technology Bachelor of Technology in Instrumentation and Control Engineering, CGPA: 6.4/10	Delhi, India Aug 2024 – Aug 2028
D.A.V. Centenary Public School, Narela Class XII – 85% Class X – 76%	Delhi, India Apr 2023 Apr 2021

Experience

Delhi Technological University Research Intern	Remote June 2025 – Aug 2025
- Conducted research on image enhancement using deep learning and classical mathematical techniques, including LCNet, Zero-DCE, CNNs, and Transformer-based approaches. - Contributed to gap analysis, experimentation planning, and study of super-resolution and image restoration methods.	
OpenCV Open Source Contributor	Remote Dec 2025 – Present
- Contributed to OpenCV by fixing JavaScript bindings for namespaced <code>Ptr<T></code> factory method return types. - Submitted pull requests and collaborated with maintainers on debugging, testing, and improving OpenCV JS bindings. - Worked with CMake, Emscripten, and the OpenCV JavaScript build pipeline.	
IEEE NSUT Joint Secretary	Delhi, India Apr 2024 – Present
- Led organization of multiple AI/ML and engineering workshops, managing end-to-end execution. - Coordinated with speakers, sponsors, and cross-functional teams to ensure seamless event delivery. - Managed logistics, registrations, and on-ground operations, improving participation and engagement. - Drove outreach and promotions, increasing visibility across the student community.	

Projects

Fluentiq – RAG-Powered Document Q&A System - Built a RAG-based document Q&A system using LangChain and FAISS, reducing query latency by 50%. - Designed efficient chunking and vector search pipeline for scalable document retrieval. - Tech Stack: Python, LangChain, FAISS, Streamlit, Google Gemini API.
Gravitational Lens Detection using Deep Learning - Developed a CNN-based pipeline to detect strong gravitational lenses from astronomical image datasets. - Addressed class imbalance using focal loss and data augmentation techniques. - Evaluated models using ROC-AUC, precision-recall metrics, and Grad-CAM interpretability. - Built using Python, PyTorch, NumPy, and computer vision techniques.
F1 Constructor Performance Predictor - Developed an ML web app to predict constructor points using historical race data. - Engineered 15+ features and achieved $R^2 > 0.85$. - Tech Stack: Python, Scikit-learn, Pandas, Plotly, Streamlit.
IEEE NSUT Website Development - Contributed to the Alumni section of the IEEE NSUT website using React and Vite. - Built responsive UI components for structured alumni profiles with smooth navigation across devices. - Tech Stack: React, Vite, Tailwind CSS, JavaScript.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL
ML/AI: TensorFlow, Scikit-learn, LangChain, FAISS, RAG, NLP, Feature Engineering, Pandas
Backend & App Development: Flask, REST APIs, MongoDB, Docker, Flutter, Streamlit
Tools: Git, GitHub, CMake, Emscripten

Leadership & Extracurricular Activities

Design, Event Management Coordinator – Innovision, Resonanz & Moksha - Managed event logistics, scheduling, venue coordination, promotions, and participant registrations. - Coordinated with design, marketing, and technical teams to ensure timely delivery and collaboration.
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Achievements & Certifications

- Secured **3rd Place** in Machine Learning Bootcamp conducted by IEEE NSUT.
- Completed **Machine Learning Specialization** by Andrew Ng on Coursera.
- Completed **Data Analytics Virtual Internship** by Deloitte Australia on Forage.